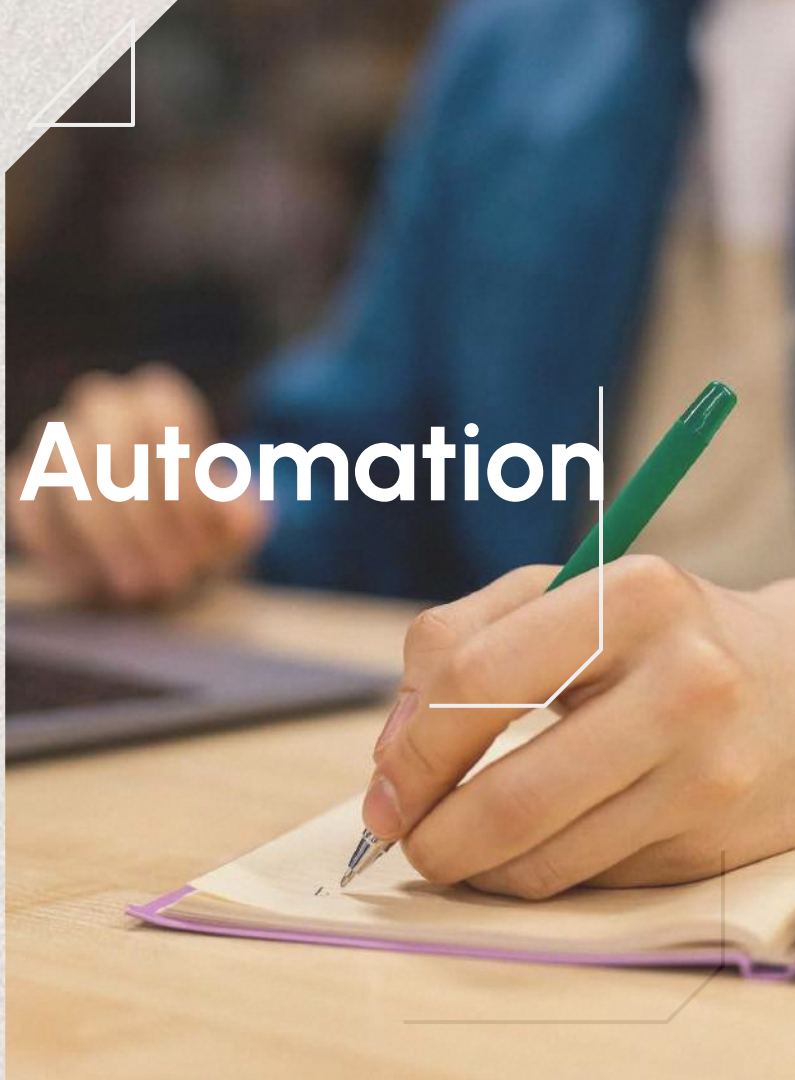
Software Quality Assurance

Introduction to Automation



Test Automation - Exercise

01 What is Test Tools

02 Test Tools Classification

03 Test Automation

04 Automation Frameworks

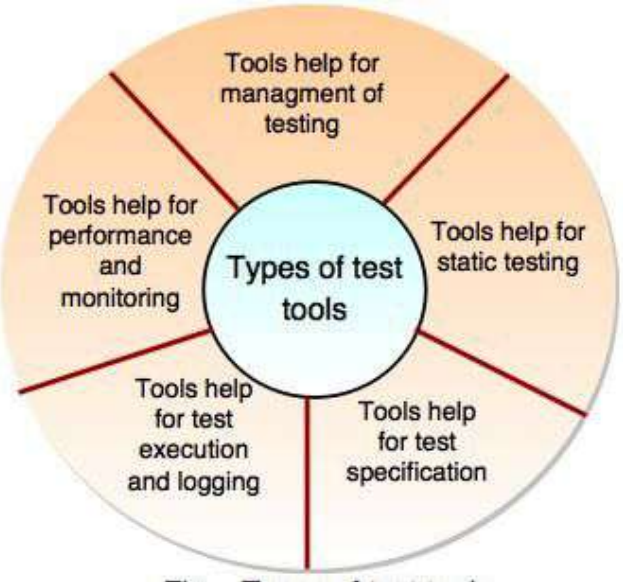
What is a Test Tool?

- A Software product that support one or more test activities, such as
  - Planning and control of test
  - Test Specifications
  - Building files and data used for test
  - Test execution
  - Test result analysis

Anything that makes testing ***faster, easier, more accurate***

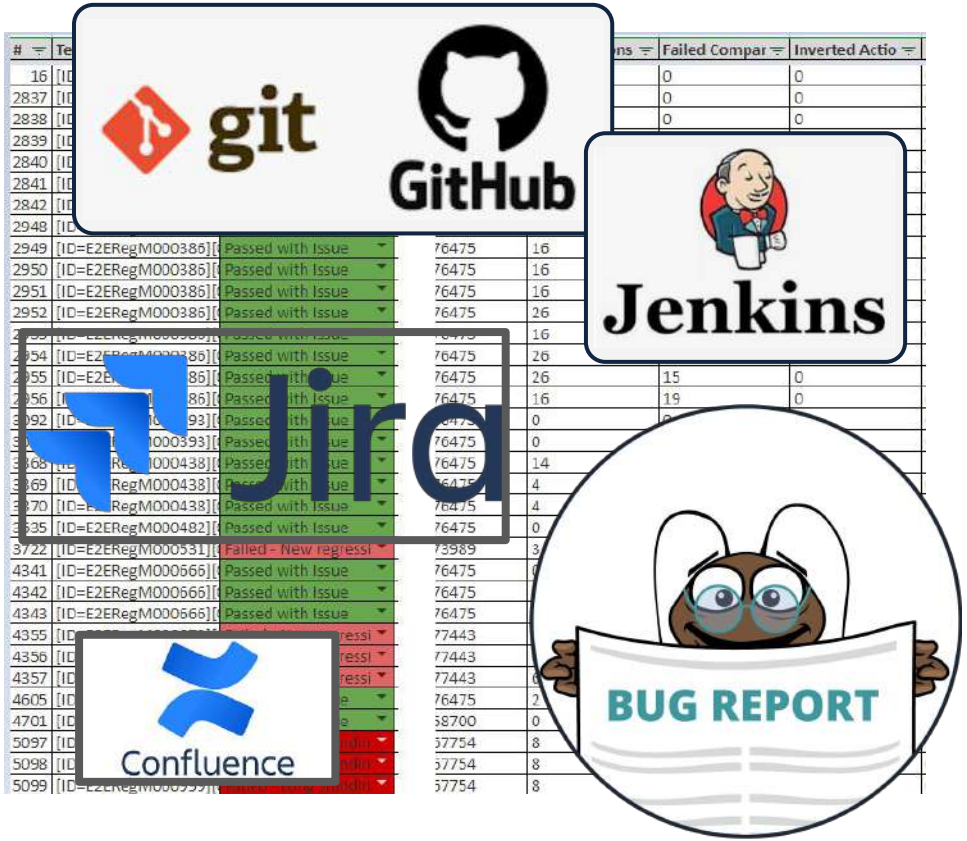
Test tool classification

- Automate repetitive and resource demanding tasks
- Automate tasks which cannot be run manually
- Support manual activities
- Increase the reliability of testing
- Improve consistent testing



Tool support for test management

- Test management tools and ALM (Application Lifecycle Management)
- Requirements management tools
- Defect management tools
- Configuration management tools
- Continuous integration tools (CI/CD)



Tool support for static testing

- Identify anomalies or defects in the code
- Analyze structures and dependencies
- Enforce coding standards

IntelliJ IDEA
JETBRAINS IDE

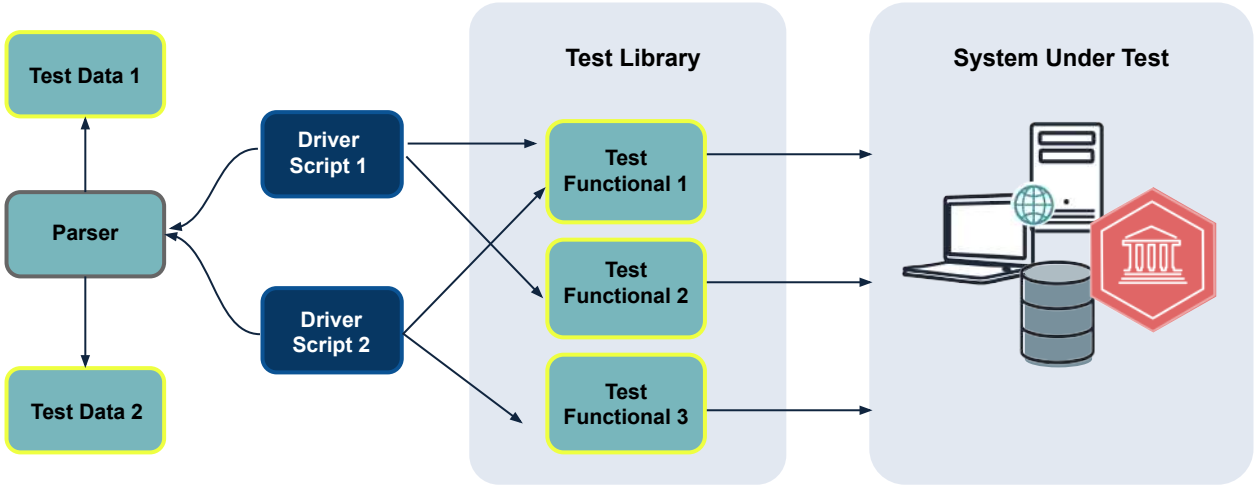
eclipse

PyCharm

Test design and implementation

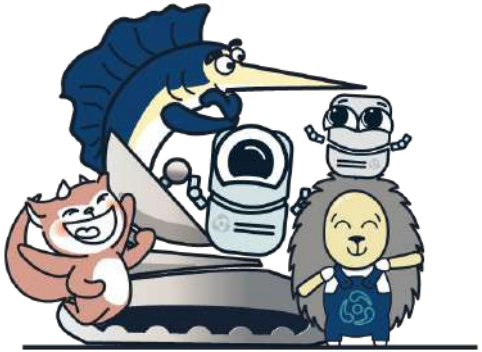
Model-based tools

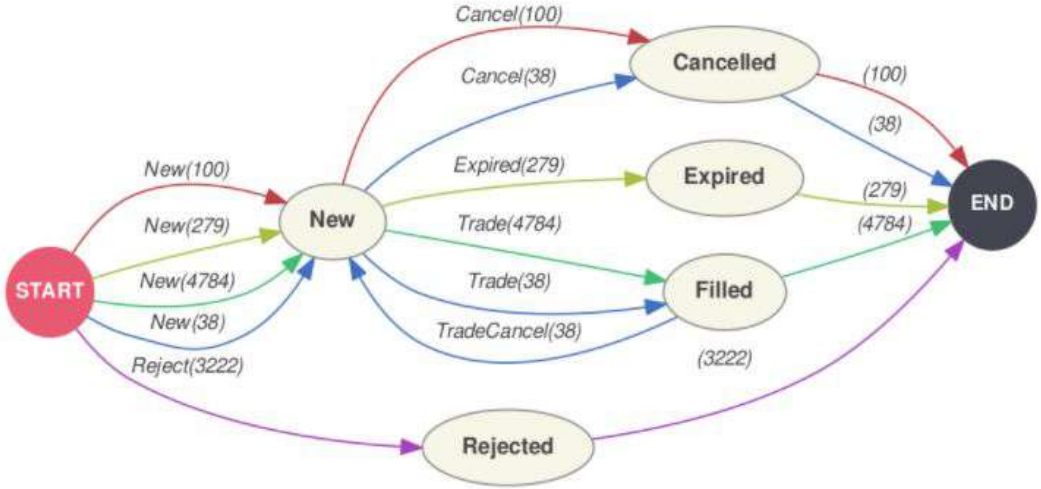
Test data preparation tools



Test execution, logging and coverage

- Test execution tools
- Coverage tools
- Test harness





Performance management and dynamic tools

- Performance testing tools
- Dynamic analysis tools

LOAD RUNNER

MiniRobots
TESTING TOOL

LoadInjector
TESTING TOOL

APACHE JMeter™

Benefits and risks of Test Automation

- Unrealistic expectations
- Under-estimated time and cost
- Tool may be relied on too much
- Open source project may be suspended

- Reduction in manual repetitive work
- Greater consistency
- More objective assessment
- Statistics about testing

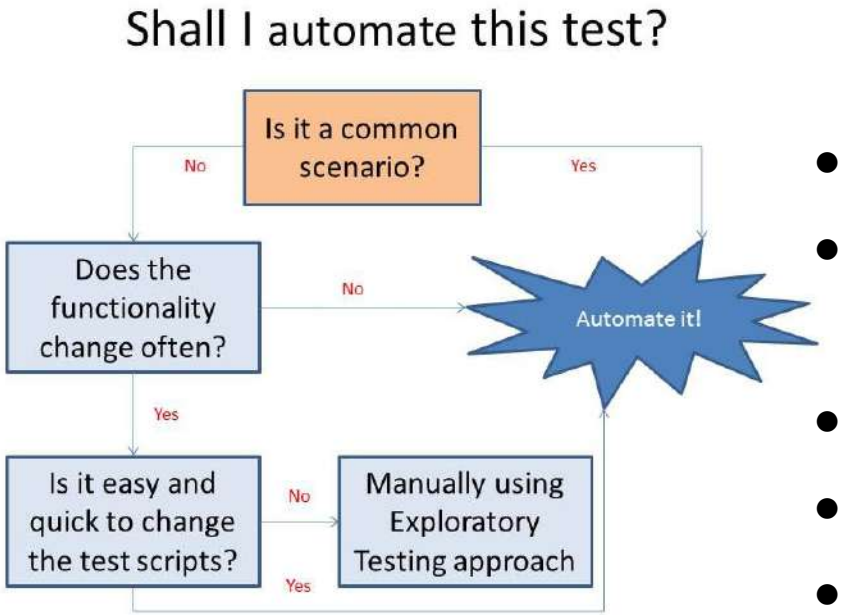
Test Automation

- Test automation is the practice of running tests automatically, managing test data, and utilizing results to improve software quality
- It’s primarily a quality assurance measure, but its activities involve the commitment of the entire software production team
- Manual process which is already in place will be automated and formalized

Test Automation Benefits

- Save time and money
- Increase test coverage
- Testing accuracy is improved – no human errors
- Can do what manual testing cannot do – large load
- Helps testers and developers
- Integration with CI/CD
- Accuracy in test reporting
- Improve QA and Dev team moral
- Regression test suite

What to Automate



- Repeatable (Set up the test including data and env, Execute the function and measure the result, Clean up the data and env.)
- Determinant (function outcome is known)
- Un opinionated (not a matter of opinion ex: Usability tests)
- High volume and load
- Role based tests
- Cross browser/cross environment tests

Test script includes

- Starting environment or processes
- Test Data creation (from API or UI)
- Test Preconditions
- Test Steps
- Verification points
- Expected outcome
- Test data clean up
- Disconnect the connections

Test Automation Tools

- Selenium

- AutoIT

- Appium

- IBM Rational Functional Tester

- UTP (QTP by HP)

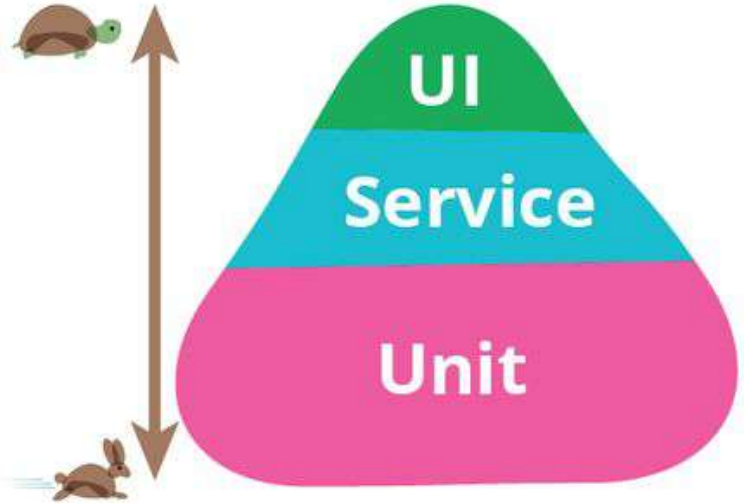
- th2

- SoapUI

- Playwright

- Cypress

Key Automation Testing Types



Unit Testing

Unit testing is the smallest testable parts of an application, called units, are individually scrutinized for proper operation. The main objective of unit testing is to isolate written code to test and determine if it works as intended

Service/API Testing

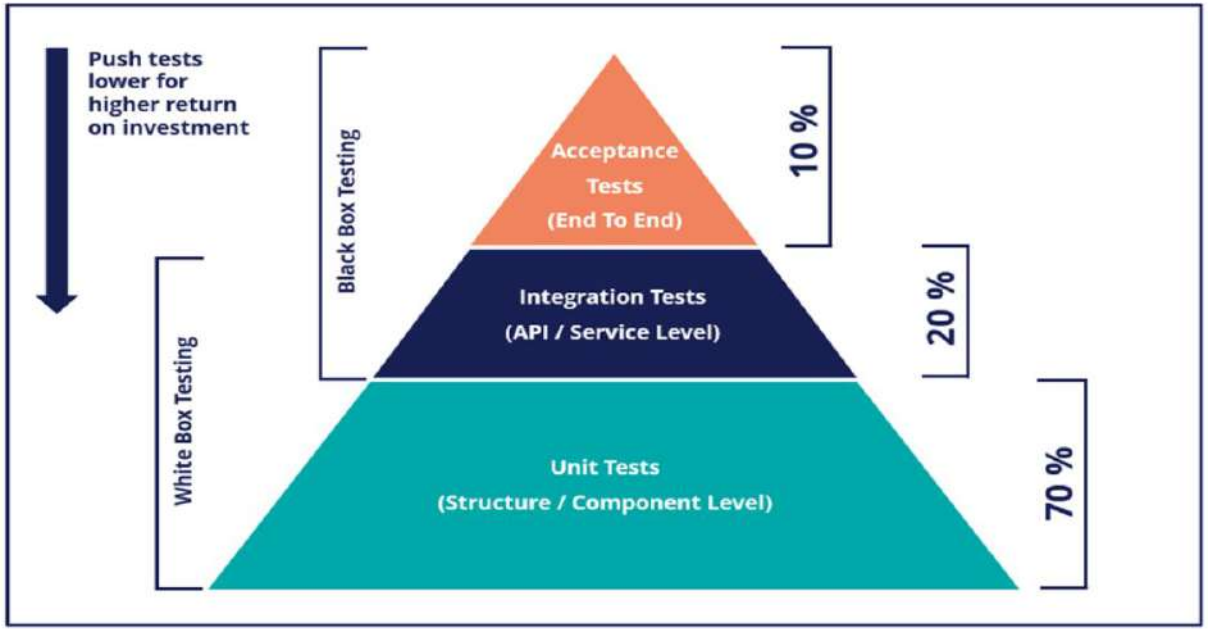
API Testing is a software testing type that validates Application Programming Interfaces (APIs)

UI Testing

The process of testing the UI of an application to verify its functionality is known as UI testing. It covers everything right from verifying the smallest bit of logic, UI workflows, navigation, transitions, calculations, validating all the buttons, etc.

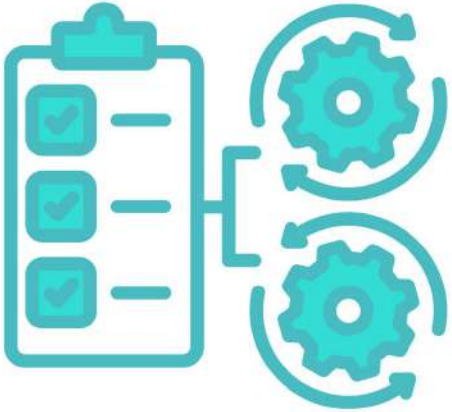
Test Automation Strategy

- Test Pyramid Strategy



Test Automation Framework

- A testing framework is a set of guidelines or rules used for creating and designing test cases.
- A combination of practices and tools that are designed to help QA professionals test more efficiently.
- Practices and guidelines are not mandatory rules , but using an organized framework provide additional benefits.



Test Framework Guidelines

- Coding standards
- Test-data handling methods
- Object repositories
- Processes for storing test results
- Information on how to access external resources

Benefits of Test Automation Framework

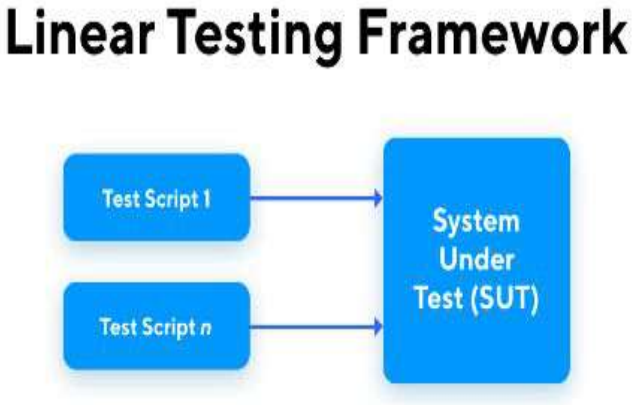
- Improve Test Efficiency
- Lower Maintenance Cost
- Minimal Manual Intervention
- Maximum Test Coverage
- Re-usability of Code

Types of Test Automation Frameworks

- Linear Automation Framework
- Modular Based Automation Framework
- Library Architecture Testing Framework
- Data – Driven Framework
- Keyword Driven Framework
- Hybrid Testing Framework

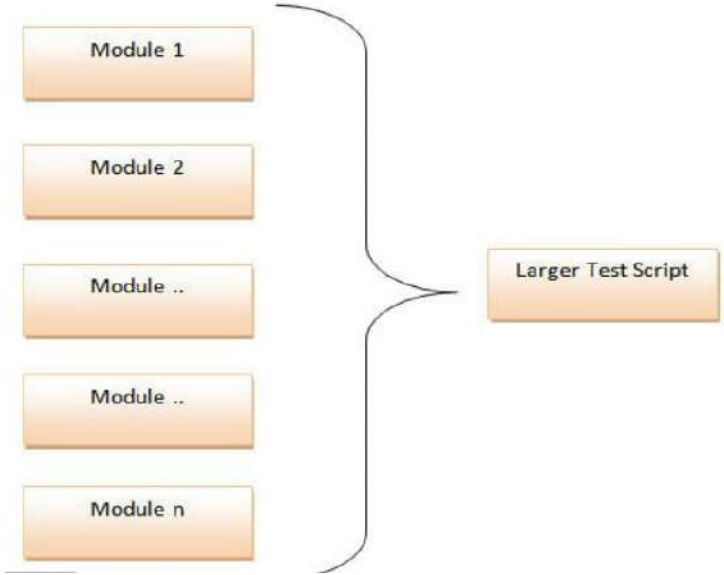
Linear Automation Framework

- Target a specific functionality.
- Create test script in sequential order.
- Run the scripts individually.
- Includes capturing all tests like navigation , inputs etc.



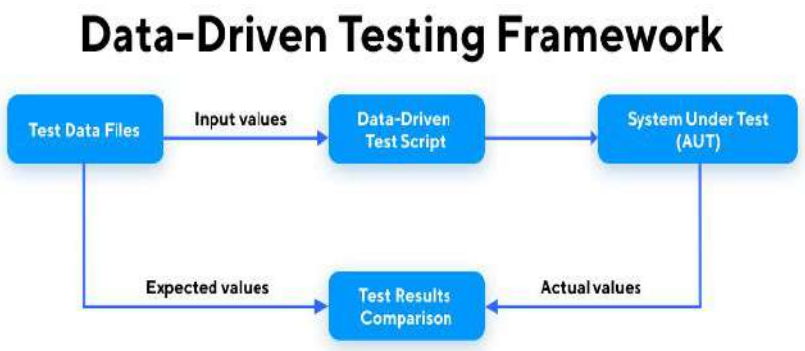
Modular Driven Automation Framework

- Modular approach.
- Breaks down test in to units, functions or modules which will be tested in isolation.
- Separate test scripts can be combined to cover complete application functionality.



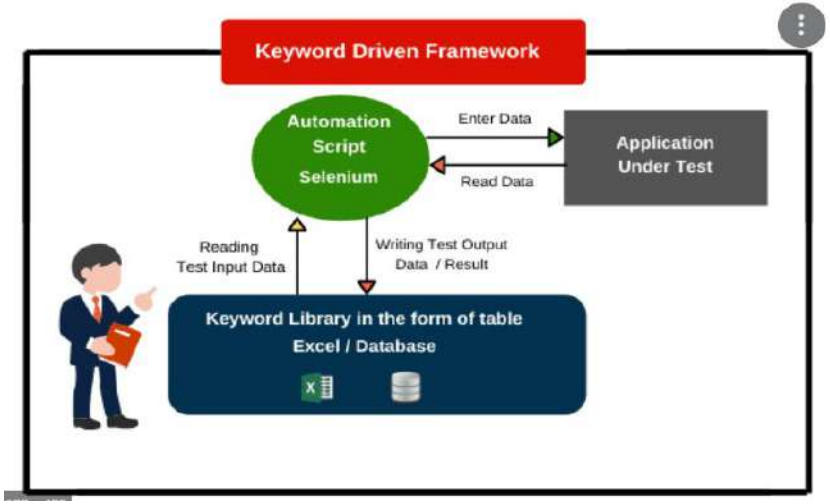
Data Driven Automation Framework

- Decouples data from the script logic.
- Ideal framework when users need to test a function or scenario with different data sets but still use the same internal logic.
- Values such as inputs and outputs are passed as parameters to test scripts from external data sources such as variable files, databases, etc.



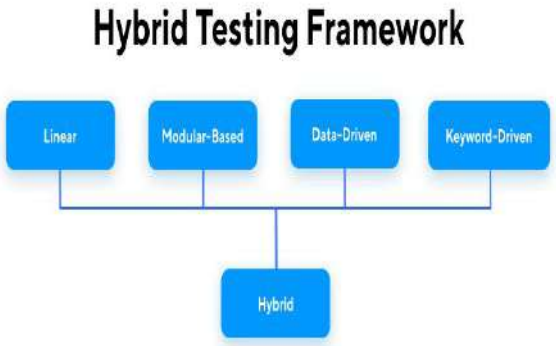
Keyword Driven Automation Framework

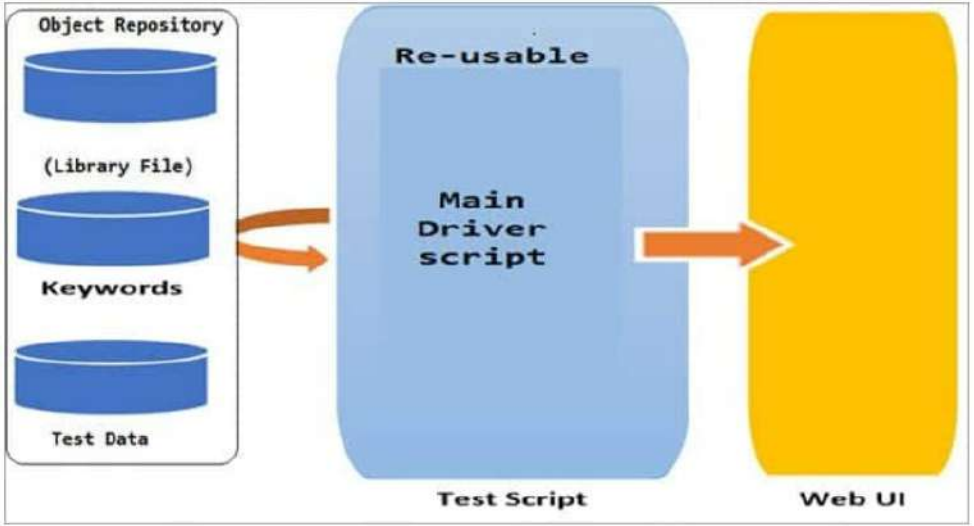
- Decoupling of data and the logic introduced in the data-driven framework a step further.
- In addition to the data being stored externally, specific keywords that are associated with different actions and used to test the GUI are also stored externally to be referenced at the test execution.
- It makes keywords independent entities that reference specific functions or actions that are associated with specific objects.



Hybrid Automation Framework

- Different types of frameworks are combined in most development environments to best suit the application testing needs.





Popular Automation Frameworks by Tech Stack

Framework	Language	Notes
Selenium	Java, Python, C#, JS	Most widely used; supports all major browsers
Playwright	TypeScript, JS, Python, .NET, Java	Fast, modern, supports multiple browsers and mobile emulation
Cypress	JavaScript	Frontend testing; great for modern web apps
Puppeteer	JavaScript	Headless Chrome automation; ideal for web scraping/testing
WebDriverIO	JavaScript	Selenium-based; supports BDD & visual testing plugins

Emerging Trends in Test Automation

API and Services Test Automation

Decoupling the client and server is a current trend in designing both Web and mobile applications.

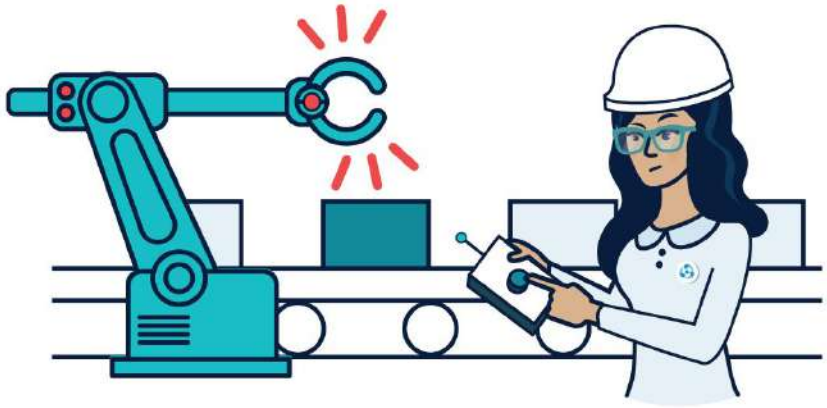
Scriptless Automation (No / Low Code)

These tests will be built using

- Model
- Keyword
- object
- Data-based approaches

Scriptless test automation enables testers to rapidly create automated tests around the functionalities of the software application with little coding knowledge.

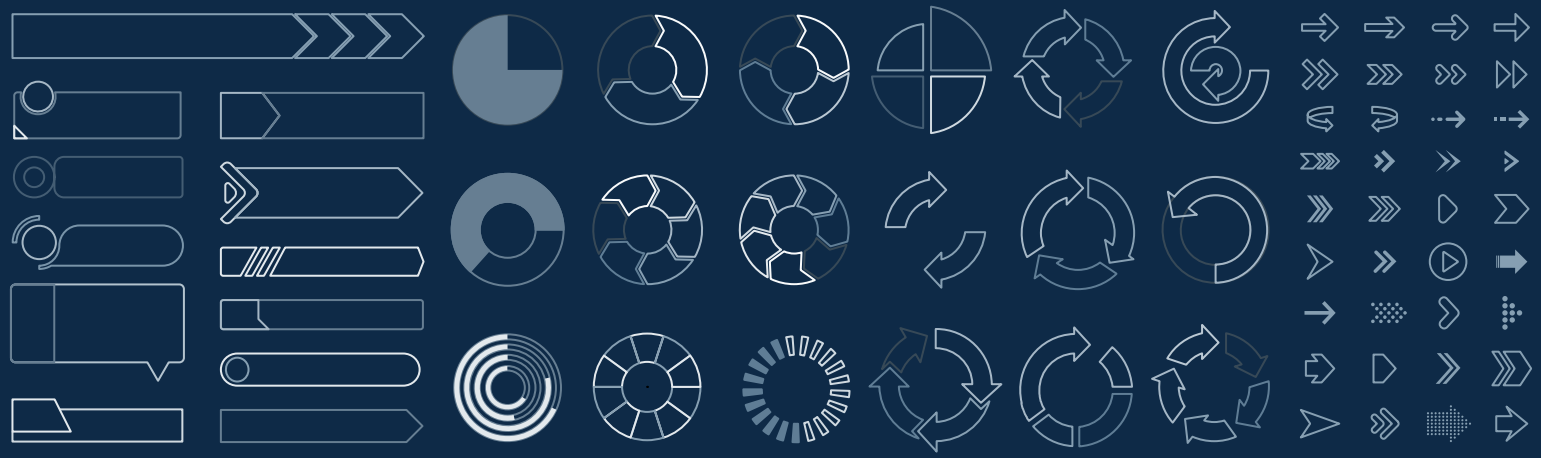
tosca ACCELQ th2 Accello DTA



AI and ML in Test Automation

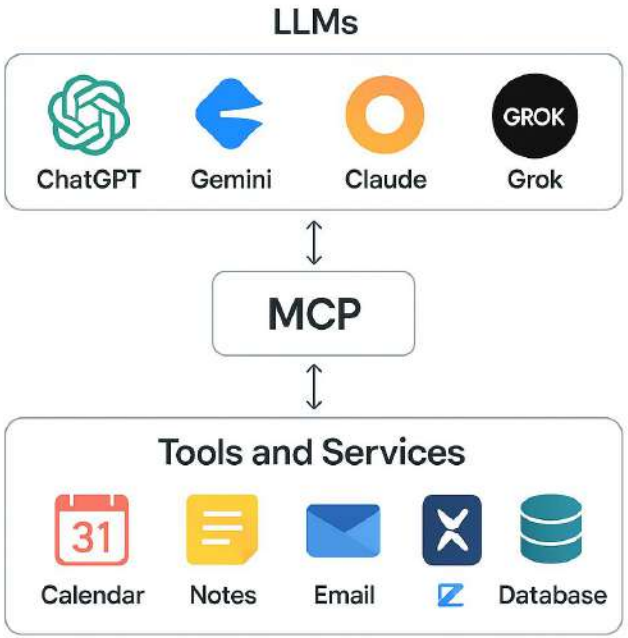
Leveraging Artificial Intelligence and machine learning to enhance and optimize test automation, including test case generation, test script maintenance , test execution and result prediction etc.

Demo



Model Context Protocol (MCP)

The Model Context Protocol (MCP) is an open standard and open-source framework introduced by Anthropic to standardize how AI systems, particularly large language models (LLMs), interact with external tools and data sources.



Questions ?

Thanks !